

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1281

Novato 138 kV
December 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1281. The Interconnection Customer proposes a 286.5 MW solar photovoltaic facility interconnecting at the Novato 138 kV substation in the West Virginia area. The request was received on September 03, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 439 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,876,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rio Grande City - Novato 138 kV
Contingency: Tres Vias - Novato 138 kV
Loading: 167.9% (Rating: 210 MVA)
DFAX: 55.6%
Pre-project loading: 92.0%
2. Facility: Rio Grande City - Millbrae 138 kV
Contingency: Tres Vias - Novato 138 kV
Loading: 167.6% (Rating: 208 MVA)
DFAX: 55.6%
Pre-project loading: 90.9%
3. Facility: Novato - Tres Vias 138 kV Line
Contingency: Rio Grande City - Novato 138 kV Line
Loading: 147.9% (Rating: 293 MVA)
DFAX: 67.5%
Pre-project loading: 81.9%
4. Facility: Firstenergy Harrison Power Sta - Tres Vias 138 kV Line
Contingency: Novato - Rio Grande City 138 kV Line
Loading: 129.7% (Rating: 266 MVA)
DFAX: 47.1%
Pre-project loading: 79.0%
5. Facility: Harrison - Firstenergy Harrison Power Station 138 kV L
Contingency: Novao - Rio Grande City 138 kV
Loading: 124.7% (Rating: 271 MVA)
DFAX: 47.1%
Pre-project loading: 74.9%

6. Facility: Morgantown Energy Facility - Lodi 138 kV
Contingency: Chester Generating Station - Millbrae 138 kV
Loading: 111.7% (Rating: 265 MVA)
DFAX: 12.3%
Pre-project loading: 98.4%
7. Facility: Millbrae - Chester Generating Station 138 kV
Contingency: Halsey - Border Steel 138 kV L
Loading: 109.6% (Rating: 343 MVA)
DFAX: 18.5%
Pre-project loading: 94.2%
8. Facility: Halsey Substation - Lodi 138 kV
Contingency: Millbrae - Chester Generating Sta 138 kV
Loading: 109.0% (Rating: 298 MVA)
DFAX: 12.3%
Pre-project loading: 97.2%
9. Facility: Halsey - Border Steel 138 kV
Contingency: Millbrae - Chester Generating Sta 138 kV
Loading: 108.6% (Rating: 303 MVA)
DFAX: 12.3%
Pre-project loading: 97.0%
10. Facility: Firstenergy Rivesville - Chester Generating Sta 138 kV
Contingency: Harrison - Ncpa 1&2 345 kV L
Loading: 106.9% (Rating: 223 MVA)
DFAX: 12.7%
Pre-project loading: 90.5%
11. Facility: Firstenergy Rvesville - Grant Town Power Plant Substation 138 kV
Contingency: Harrison - Ncpa 1&2 345 kV
Loading: 104.6% (Rating: 225 MVA)
DFAX: 12.7%
Pre-project loading: 88.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Novato - Rio Grande City 138 kV	Rebuild line with higher-capacity conductor	\$7,680,000	92% (\$7,066,000)
Millbrae - Rio Grande City 138 kV	Rebuild line with higher-capacity conductor	\$9,583,000	92% (\$8,817,000)
Novato - Tres Vias 138 kV Line	Rebuild line with higher-capacity conductor	\$2,356,000	82% (\$1,936,000)
Firstenergy Harrison Power Station - Tres Vias 138 kV Line	Rebuild line with higher-capacity conductor	\$3,865,000	84% (\$3,257,000)
Harrison - Firstenergy Harrison Power Sta Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,106,000	84% (\$2,617,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Morgantown Energy Facility - Lodi 138 kV	Rebuild line with higher-capacity conductor	\$1,234,000	37% (\$454,000)
Millbrae - Chester Generating Station Substation 138 kV	Reconductor line	\$3,793,000	50% (\$1,886,000)
Halsey Substation - Lodi 138 kV	Reconductor line	\$4,484,000	37% (\$1,649,000)
Halsey Substation - Border Steel 138 kV	Reconductor line	\$2,208,000	37% (\$812,000)
Firstenergy Rivesville - Chester Generating Sta 138 kV	Reconductor line	\$1,053,000	53% (\$558,000)
Firstenergy Rivesville - Grant Town Power Plant 138 kV	Reconductor line	\$1,554,000	53% (\$824,000)
	Total allocated to project		\$29,876,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.